HealthConnect Rwanda: Analytics Testing and Refinement Report

1. Introduction

In Week 6, I worked on the HealthConnect Rwanda project, where I used PostgreSQL, SQL and Power BI to analyse healthcare data and build a seven-page interactive dashboard.

The dashboard contained important information about patients, hospitals, revenue, insurance claims, staff and patient satisfaction. I also wrote 40 SQL queries to answer different business questions and identify areas that needed improvement.

Some of the major findings from Week 6 were:

- Total billed revenue was RF9.31 billion.
- About 90% of the billed revenue had been collected.
- Only about 50% of insurance claims were approved.
- The average patient satisfaction score was 2.99.
- Hospital HF014 had a high number of appointments but relatively low patient satisfaction.

However, I also noticed some problems with the data and dashboard. For example, 55,000 out of 60,000 bills and diagnoses could not be linked to appointment records. Some dashboard visuals were not displaying information correctly, and the dataset did not contain an attendance or no-show field.

The main aim of Week 7 was to test the dashboard, check whether the figures were correct, identify and fix display problems, and make sure the conclusions from Week 6 were supported by the data.

I also prepared the findings to be shared with the Project Management and Data Science tracks.

2. How I Tested the Dashboard

I used four main methods to test the dashboard and check the accuracy of the results.

1. Checking the calculations

I recalculated the major KPIs using the original CSV files instead of relying only on the SQL queries or Power BI dashboard. I then compared the results to make sure the figures matched.

2. Checking the dashboard visuals

I went through each dashboard page to check whether the charts, tables and cards were displaying information correctly. I looked out for missing values, incorrect rankings, repeated figures and numbers that were cut off.

3. Testing dashboard interactions

I tested the slicers, drill-through feature, tooltips and bookmarks to make sure they worked properly and displayed the expected results when selected.

4. Checking results across different groups

I checked some of the major findings across different hospitals, insurance companies and months. This helped me understand whether the results were consistent or whether they were influenced by a particular group.

I also kept screenshots of the dashboard before and after making corrections as evidence of the work carried out.

3. Checking the Accuracy of the KPIs

One of the most important parts of this week's work was confirming that the figures on the dashboard were correct.

I compared the dashboard values with the results calculated from the original data.

KPI	Correct value from source data	Dashboard value	Result
Total patients	20,000	20K	Passed
Total appointments	5,000	5K	Passed
Total revenue	RF9,309,401,961	RF9.31bn	Passed
Total collected	RF8,389,454,556	RF8.39bn	Passed
Outstanding amount	RF919,947,405	RF919.9M	Passed
Collection rate	90.12%	90%	Passed
Claim approval rate	49.98%	50%	Passed after retesting
Average satisfaction	2.99	2.99	Passed
Poor ratings	40.44%	40%	Passed
Emergency appointments	1,650 (33%)	1,650 (33%)	Passed
Average patient age	45.03	45	Passed
Patients with chronic conditions	14,929	15K	Passed
Staff, beds and equipment	1,250; 2,000; 200	Same values	Passed

I also checked the hospital and department revenue rankings. The top three hospitals were HF011, HF006 and HF013, while Cardiology, Radiology and Laboratory were the top three departments.

All the KPI calculations I checked matched the original data. This means I did not find any calculation errors in the main dashboard measures.

4. Dashboard Testing and Corrections

During the first round of testing, I discovered seven problems with the dashboard. These were mainly related to how information was displayed, rather than errors in the actual calculations.

After making corrections, I tested the dashboard again and discovered two additional issues. I fixed those as well.

Below are some of the major problems I found and how I corrected them.

1. Hospital information was showing the same figures

Initially, every hospital was displaying the same number of appointments, staff, beds and equipment. This was because the wrong field was being used to connect the information in the dashboard.

I corrected the field so that each hospital would display its own information. After the correction, HF001, for example, showed 273 appointments, 64 staff members, 100 beds and 10 equipment items.

2. Bed information was incorrect

The bed chart was showing 2,000 beds for every hospital instead of showing the correct number for each facility.

I corrected the field used in the visual, and each hospital now displays approximately 100 beds.

3. A blank row appeared in the hospital table

During another test, I noticed a blank hospital row showing 100% bed utilisation.

I adjusted the Bed Utilization percentage measure so that it would return a blank value when there were no beds to calculate. This removed the unwanted row.

4. Department rankings were not properly arranged

The department satisfaction chart was displaying departments in alphabetical order instead of ranking them by satisfaction.

I changed the sorting to show departments from the highest satisfaction score to the lowest. Emergency came first with 3.02, while ICU had the lowest score of 2.76.

5. Blank department category

A blank department category was appearing in the rankings and decomposition tree.

I removed the blank category so that only the available departments were displayed.

6. Revenue figure was cut off

The revenue card was displaying an incomplete figure as RF9.3 instead of showing the full amount.

I increased the card size and adjusted the display settings. It now shows RF9.31 billion clearly.

7. Monthly revenue chart was incomplete

The monthly revenue chart was not showing all 12 months. It stopped around September.

I adjusted the chart size, and all 12 months are now visible.

8. Rwanda map was not displaying properly

The province map was not clearly focused on Rwanda.

I adjusted the visual so that patient numbers across the provinces could be viewed more clearly.

9. Monthly growth table had an incorrect total

The total row in the month-over-month growth table was comparing different periods in a way that could be misleading.

I removed the total row so that the monthly growth percentages could be viewed correctly.

10. Some KPI cards were showing incorrect display values

During retesting, I noticed that some cards were displaying 0K appointments for HF011 and 0.07bn revenue for March.

I corrected the display settings so that the actual figures would appear. HF011 now shows 273 appointments and RF44.21 million in revenue. March shows 440 appointments and RF69.17 million.

11. Patient page had an unwanted filter selection

The patient page was initially opening with the Eastern province selected, showing only 4,067 patients instead of the full 20,000.

I cleared the filter so that the page now displays all patients when opened.

Testing the Dashboard Interactions

After correcting the visuals, I also tested the interactive features to make sure they worked properly.

The results were:

- Selecting HF011 displayed 273 appointments and approximately RF44.2 million in revenue.
- Selecting March displayed 440 appointments.
- Selecting Female displayed 9,910 patients.
- Selecting Western province displayed 4,091 patients.
- Using drill-through for HF011 displayed its appointments, revenue, staff and satisfaction figures correctly.
- The custom tooltip appeared when hovering over a hospital chart.
- The bookmark buttons successfully switched between the chart and table views.

All these interaction tests passed after the necessary corrections.

5. Checking Whether the Findings Were Consistent

I did not want to rely on the overall dashboard figures alone. I also checked whether some of the findings remained the same when looking at different hospitals, insurance companies and months.

Some of the results were consistent.

- Insurance claim approval remained between 49.3% and 50.9% across all four insurance companies. This shows that the approval problem was not limited to one insurer.
- Emergency appointments remained between 30.1% and 36.5% every month. This supports the overall finding that about 33% of appointments were emergencies.
- The revenue rankings for hospitals and departments also matched the original data.

However, some findings needed to be treated carefully.

For hospital satisfaction, the number of linked feedback records was relatively small, ranging from 72 to 102 records per hospital. Poor ratings also varied between 34.1% and 48.5%.

This means that differences between hospitals may not tell the full story.

For example, HF014 had 276 appointments and an average satisfaction score of 2.77. This makes it worth investigating, but the available feedback records are not enough to conclude that it is definitely one of the worst-performing hospitals.

Department satisfaction also had limited feedback records, ranging from 146 to 188 per department. ICU had the lowest score of 2.76, while the other departments ranged from 2.88 to 3.02.

6. Important Findings After Testing

After checking the calculations, correcting the dashboard and reviewing the results, I identified five important findings.

1. Bill status does not always show the correct payment situation.

The data showed that collection rates were almost the same across different bill statuses. Paid bills had a 90.0% collection rate, partially paid bills had 90.1%, and pending bills had 90.2%.

I also found that 4,042 bills marked as Paid still had unpaid balances, amounting to RF311.2 million.

This means that the hospital should not depend only on the bill status when checking payments. It should also look at the actual amount billed, amount paid and outstanding balance.

2. Patient comments do not always match their ratings.

I discovered that 2,078 out of 5,283 comments labelled as Poor actually had ratings of 4 or 5 stars.

This shows that some written comments do not agree with the ratings given by patients. Therefore, the numerical ratings are more reliable for measuring satisfaction in this dataset.

3. Some appointment dates are in the future.

I found 821 appointments with dates after 29 October 2026, extending up to 28 December 2026.

These dates need to be checked to confirm whether they are correct or whether there was a problem when the data was entered.

4. There is a major problem with linking records.

Only 5,000 out of 60,000 bills and diagnoses could be linked to appointment details. Also, only 1,756 out of 21,064 feedback records were linked to appointments.

This makes it difficult to connect different hospital activities and understand the complete patient journey.

5. The dataset does not contain attendance information.

There is no field showing whether a patient attended an appointment or failed to show up.

Because of this, I could not analyse attendance or no-show patterns. The closest available information was the visit type, which included Emergency, Follow-up and Consultation.

7. Updated Recommendations

Based on the findings from Week 7, I have updated some of my earlier recommendations.

1. Improve payment collection and follow up on outstanding bills.

About RF920 million is still outstanding. Hospitals should review actual payment amounts instead of depending only on bill status because some bills marked as Paid still have unpaid balances.

2. Review the insurance claims process.

Since claim approval is around 50% across all four insurers, the hospitals should investigate why so many claims are not being approved.

3. Improve how staff are distributed across hospitals.

Some hospitals receive more appointments than others, but their staffing levels are almost the same. Hospitals should consider patient demand when planning staff allocation.

4. Improve patient satisfaction.

The ICU had the lowest department satisfaction score, so it should be given attention. HF014 should also be investigated further, although more patient feedback is needed before making a final judgement.

5. Pay attention to older patients and those with chronic conditions.

The data showed that about 75% of patients had chronic conditions. Hospitals should continue to pay attention to the needs of these patients.

6. Reduce the number of emergency visits where possible.

Emergency appointments made up about 33% of all appointments throughout the year. Hospitals should investigate the reasons behind this and consider ways to improve early access to healthcare.

7. Improve data collection and record keeping.

Hospitals should ensure that bills, diagnoses, feedback and appointments can be properly linked. They should also record patient attendance, validate appointment dates and include actual hospital names instead of using only hospital IDs.

8. How the Findings Can Help Other Tracks

The findings from this project can also be useful to the other teams working on HealthConnect Rwanda.

Data Science Track

The dataset does not contain patient attendance information. Therefore, it is not possible to build a reliable model for predicting missed appointments using the available data.

Also, patient comments should not be used as the main measure of satisfaction because some comments do not match the ratings given.

The small number of feedback records for individual hospitals should also be considered when building any model.

Project Management Track

The project team should pay attention to the data problems identified during testing.

These include appointment dates appearing in the future, bill statuses that do not always match payment records, and the absence of attendance information.

These issues should be properly documented and followed up so that they can be corrected.

9. Cross-Track Contribution

One of the requirements for this week was to carry out a real test with someone from another HC-POD track.

Unfortunately, I was unable to complete this part because no teammate was available at the time of writing this report.

However, I prepared a summary of the validated findings to share with the Project Management and Data Science teams.

The plan is to share the findings with the other tracks, allow them to review and either agree or disagree with the conclusions, record any issues they identify, make necessary corrections and test the results again.

This activity is still outstanding and will be completed if a teammate becomes available before Week 8.

10. Limitations and Remaining Issues

Although the dashboard has been tested and improved, there are still some limitations in the available data.

- Some hospital, department and monthly results are based only on records that could be linked, while the overall figures use all available records.
- Hospitals are identified using codes such as HF001 and HF014 instead of their actual names and locations.
- All 2,000 beds in the dataset are recorded as available. This makes bed utilisation appear as 0%, but it does not necessarily mean that the beds are actually empty.
- The dataset does not contain laboratory, prescription, admission, referral or patient attendance information.
- The currency is assumed to be Rwandan francs (RF).
- The cross-track testing activity has not yet been completed.

These limitations are important because they affect how some of the findings should be understood and used.

11. Plans for Week 8

For Week 8, I plan to focus on the following activities:

1. Complete the cross-track testing with a member of the Project Management or Data Science team and include their feedback in the report.
2. Add actual hospital names and patient attendance information if new data becomes available.
3. Investigate the appointment dates that appear to be in the future.
4. Keep the dashboard simple and include a short explanation of the data limitations on each page.
5. Prepare the final project presentation using the findings that have been checked and confirmed.

Conclusion

Week 7 gave me the opportunity to go back to the work I completed in Week 6 and check whether the dashboard and its findings were reliable.

I was able to confirm that the main KPI calculations matched the original data. I also discovered and corrected several dashboard display problems, tested the interactive features and checked whether important findings remained consistent across different groups.

One major lesson from this exercise is that having a dashboard with figures and charts is not enough. It is also important to make sure the figures are correct, the visuals are clear and the conclusions are supported by the available data.

Although some data limitations and the cross-track testing activity are still outstanding, the dashboard is now in a better state and the findings are clearer.

This work has helped me understand the importance of testing, validating and improving an analytics solution before presenting it for business decisions.